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Self-Supervised Quantification of Despeckling Uncertainties Applied to Unsupervised Radiometric Change Detection in SAR imaging

Thomas Bultingaire, Christophe Kervazo, Loïc Denis, and Florence Tupin

***Abstract*—The Synthetic Aperture Radar (SAR) imaging modality is well-suited for change detection tasks. In particular, its robustness to day-night cycles and cloud coverage is critical for certain applications, such as disaster assessment and monitoring the deforestation in old-growth forests that are frequently cloud-covered. However, its use is often impeded by the speckle phenomenon, causing strong fluctuations in the radar images, which are difficult to distinguish from actual temporal changes. To suppress these fluctuations, we propose to resort to a state-of-art despeckling method. In addition, to better take into account the residual fluctuations in the areas that are the most challenging to restore, we quantify despeckling uncertainties, which enables reaching a constant rate of false positive detections. We also show that incorporating spatial correlations into the despeckling uncertainties model can further improve change detection, especially in highly textured regions such as building areas. The proposed SAR Despeckling Uncertainty Change detection frameworK (SAR-DUCK) is applied to TerraSAR-X and Sentinel-1 images, demonstrating improved detection rates compared to methods from the state-of-the-art, as well as a constant and controlled false alarm rate. Supplementary results and an implementation of the proposed method are available on GitLab at https://gitlab.telecom-paris.fr/ring/sar-duck.**

***Index Terms*—Unsupervised change detection, despeckling uncertainty quantification, SAR imaging.**

I. Introduction

CHANGE detection in remote sensing aims to detect and localize changes on Earth's surface between two dates. Generally speaking, change detection is studied across many imaging modalities [1] and has a wide range of applications, such as disaster assessment (after floods, earthquakes, volcanic eruptions, military actions, *etc.*) or the monitoring of ice, forests, or agriculture [2]. Change detection can be divided into *semantic change detection*, aiming to both detect and classify changes [3], [4]; and (binary) *change detection*, either focusing on a specific type of change (*e.g.*, urban changes [5], deforestation [6], crop classification [7]), or listing all visible changes between two images [8], [9] (referred to, in the following, as *radiometric change detection*). Depending on the imaging modality and the task, change detection is approached in most recent works with supervised [5], [10], [11], weakly-supervised [12], or unsupervised deep-learning [13], [14]. In this paper, we focus on the latter.

The robustness of Synthetic Aperture Radar (SAR) imaging to the day-night cycle and cloud coverage makes it especially suitable for change detection. This is particularly crucial for disaster assessment, to act as quickly as possible regardless of the weather, and for deforestation monitoring in areas such as Amazonia, which are frequently cloud-covered. In this work, we focus on unsupervised *radiometric change detection* in SAR imaging. In addition to being application-specific, producing a ground truth labeling of image pairs for change detection is particularly tedious and time-consuming, hence the interest in unsupervised methods. In this context, traditional approaches are based on thresholding or clustering of difference images [8], [15], [16]. More recently, deep-learning techniques have driven change detection research. The self-supervised construction of a suitable representation space via contrastive learning has been proposed in [9], followed by a thresholding or clustering step. Another more widely spread approach [13], [17], [18] consists of training an end-to-end change detection network using pseudo-labels generated by an unsupervised method, such as fuzzy C-means [19]. Instead of directly predicting changes, [20] trains a Siamese neural network with contrastive learning using pseudo-labels generated by FLICM [21] (derived from fuzzy C-means) applied to a difference in a local neighborhood. The use of superpixels is also studied in [22], [23] to extract richer features, which are then fed to the change detection network. Finally, pseudo-labels are refined in [24]: if a first neural network (trained with pseudo-labels) labels a pixel differently from its initial pseudo-label, then the pixel is removed from the training set and a second training is conducted.

Change detection is impeded by the presence of speckle [25] in SAR images, which represents a highly challenging issue for change detection: even if the scene reflectivity remains unchanged, due to speckle decorrelation, successive observations exhibit large intensity fluctuations. Consequently, for change detection, distinguishing true changes from fluctuations due to speckle can be challenging. To bypass this issue, most SAR change detection methods resort to multi-looked georeferenced images. While multi-looking reduces speckle fluctuations, it also strongly degrades spatial resolution. The impact of speckle can be suppressed more effectively through a learning-free despeckling step [26], a deep-learning despeckling step [27], or through object-based image analysis [28].

In this article, instead of performing speckle reduction with spatial multi-looking [29], we rather use Single-Look Complex (SLC) self-supervised despeckling techniques, al-

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The authors are with LTCI, Télécom Paris, Institut Polytechnique de Paris, Palaiseau, France.

Loïc Denis is also with the Université Jean Monnet Saint-Etienne, CNRS, Institut d Optique Graduate School, Laboratoire Hubert Curien UMR 5516, F-42023, Saint-Étienne, France.

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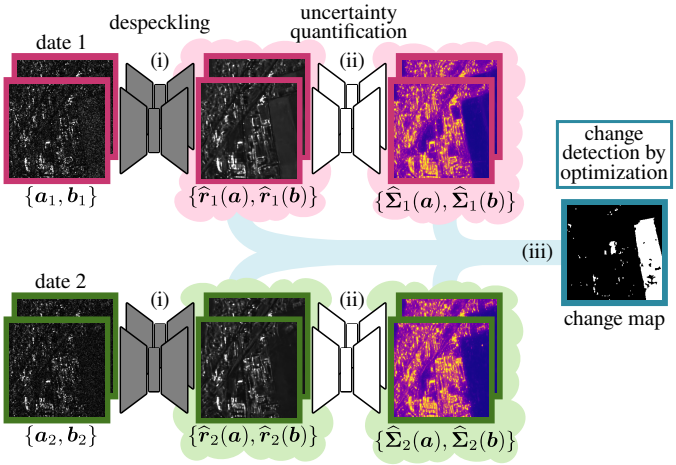


Fig. 1. Overall scheme of the proposed method SAR-DUCK. (i) First, both real and imaginary parts of a SLC SAR image are despeckled. (ii) The uncertainty of the despeckling step is then quantified using a neural network. (iii) Finally, changes between the two dates are detected through an optimization process.

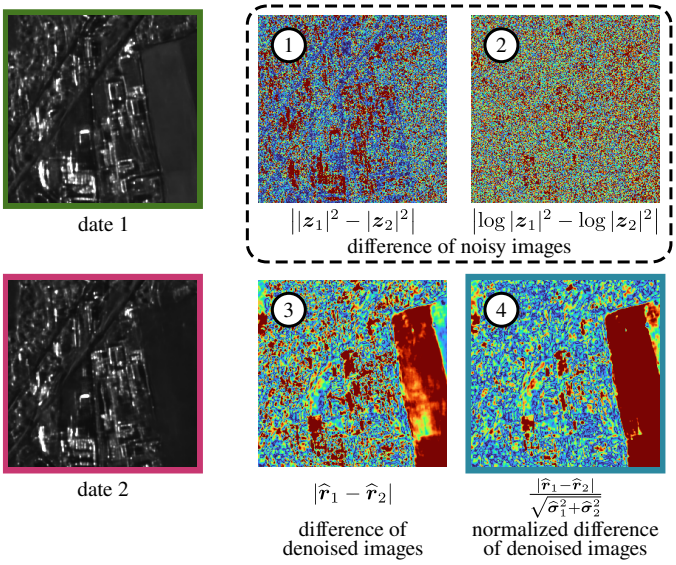


Fig. 2. Detecting changes using difference images. The difference between the noisy images (either intensities $|z_i|^2$ ① or log-intensities ②) is too noisy. The difference between despeckled log-reflectivities $\widehat{r}_i$ ③ is more adapted to change detection but leads to more frequent false alarms in regions where despeckling is less effective (*e.g.*, building areas). This issue can be addressed by normalizing the difference by the despeckling uncertainties $\widehat{\sigma}_i$ ④.

lowing speckle removal with minimal loss of resolution [30]. These methods can, for example, separately exploit the real and imaginary parts of the complex image [31]. This is the counterpart of self-supervised methods in computer vision, such as Noise2Noise [32], Noisier2Noise [33] or Recorrupted-to-Recorrupted [34]. Yet, these data-driven methods generate only an estimate of the true underlying speckle-free image. This estimate necessarily contains residual fluctuations (related to so-called aleatoric uncertainties [35], *i.e.*, uncertainties due to the speckle realization) that affect change detection. We propose to mitigate the impact of these residual fluctuations on change detection by designing and leveraging a self-supervised uncertainty quantification method. The proposed SAR Despeckling Uncertainty Change detection frameworK (SAR-DUCK) is summarized in Figure 1. The main contributions of this paper, which extend our preliminary work [36], are the following:

- a self-supervised approach to quantify despeckling uncertainties and their spatial correlations inside small patches (step (ii) in Figure 1). The uncertainty is estimated thanks to the separation of SLC images into their real and imaginary parts;
- an unsupervised radiometric change detection technique that preserves the original resolution of image pairs (step (iii) in Figure 1), thanks to the use of the above despeckling uncertainties inside small patches of SAR images. Our change detection method achieves a controlled Constant False Alarm Rate (CFAR).
- The validation of the change detection method for TerraSAR-X images, and forest monitoring in Amazonia using Sentinel-1 images.

II. Methodology

Strong fluctuations caused by the speckle phenomenon in single-look SAR images make change detection challenging, since they are hard to distinguish from the actual changes: see the noisy difference images in Figure 2, computed either in linear scale ① or in logarithmic scale[1] ②. Most of the change detection methods use multi-looked images, which have improved radiometric signal-to-speckle ratios, but at the cost of a lower spatial resolution. To avoid a resolution loss, we rather build on recent despeckling techniques. Yet, despeckling does not suppress speckle fluctuations equally well in all areas of the image. Despeckling two images corresponding to different speckle realizations of the same radar scene leads to two slightly different restorations, due to the so-called aleatoric uncertainties of the despeckling method, with larger remaining fluctuations in textured areas (*e.g.*, buildings) than in smooth areas (*e.g.*, fields). This makes changes harder to separate from residual fluctuations in the zones of high uncertainty of the despeckled image (③ in Figure 2). Thus, applying a constant threshold to the difference between two despeckled images leads to a higher false alarm rate in areas of stronger uncertainty. To reach a constant false alarm rate, we propose first to estimate despeckling uncertainties and then to account for them when detecting changes (④ in Figure 2).

A. Quantification of despeckling uncertainties

In the following, we develop a model of the despeckling uncertainties (Sec. II-A1), develop a method to estimate the parameters of the model, using the decomposition of SLC images into real and imaginary parts (Sec. II-A2), and finally, detail the network architecture we suggest for such an estimation (Sec. II-A3). Our approach to quantify uncertainties is general and applicable to most despeckling techniques (conditions for applying our framework are specifically discussed in part II-A2).

[1] generally used in SAR imaging to stabilize the variance of intensity fluctuations due to speckle [25].

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1) Model of despeckling uncertainties: A despeckling algorithm provides an estimate $\widehat{r}$ of the true log-reflectivity r_{true}. For each pixel, the mean squared error $\mathbb{E}[(\widehat{r} - r_{\text{true}})^2]$, where the expectation $\mathbb{E}$ is carried out over speckle realizations, is typically decomposed into a bias and a variance term:

$$\begin{aligned}\mathbb{E}\left[(\widehat{r} - r_{\text{true}})^2\right] &= \mathbb{E}\left[\left(\mathbb{E}[\widehat{r}] + (\widehat{r} - \mathbb{E}[\widehat{r}]) - r_{\text{true}}\right)^2\right] \\ &= \underbrace{\left(r_{\text{true}} - \mathbb{E}[\widehat{r}]\right)^2}_{\substack{\text{squared bias} \\ \text{(epistemic uncertainties)}}} + \underbrace{\text{Var}[\widehat{r}]}_{\substack{\text{variance} \\ \text{(aleatoric uncertainties)}}}. \end{aligned} \quad (1)$$

Note that the bias term does not necessarily indicate a systematic under- or over-estimation of the radiometry. Instead, it captures the loss of some details, such as low-contrasted features with limited spatial extension, or weakly contrasted textures that are usually repeatedly smoothed out by the despeckling algorithm. In the following, we focus on aleatoric uncertainties. Indeed, as shown in the following (Eq. (4)), only the aleatoric uncertainties affect the false alarm rate.

Beyond characterizing the variance of the despeckling technique, we also capture its covariance at the scale of $N \times N$ pixel patches (vectors are written in bold and matrices in bold upper-case letters). We approximate the fluctuations of the despeckled log-reflectivity with a Gaussian distribution:

$$\widehat{\boldsymbol{r}} - \mathbb{E}[\widehat{\boldsymbol{r}}] \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma}), \quad (2)$$

where the covariance matrix captures the strength and correlations of the remaining speckle fluctuations.

By comparing the estimates $\widehat{\boldsymbol{r}}_{\boldsymbol{a}}$ and $\widehat{\boldsymbol{r}}_{\boldsymbol{b}}$ computed from two independent speckle realizations of the same radar scene, we can capture information about $\boldsymbol{\Sigma}$ (aleatoric uncertainties): the zero-mean difference $\widehat{\boldsymbol{r}}_{\boldsymbol{a}} - \widehat{\boldsymbol{r}}_{\boldsymbol{b}}$ is distributed according to a centered normal distribution with a covariance that is doubled. Such independent estimates can be easily obtained from a SLC SAR image by separately processing the real ($\boldsymbol{a}^2$) and imaginary ($\boldsymbol{b}^2$) components. Indeed, under Goodman's fully developed model [25], these two components are independent. Such a decomposition has first been used to train despeckling networks (the so-called MERLIN approach [31]). Given this decomposition, the final estimate of the log-reflectivity is generally defined by:

$$\widehat{\boldsymbol{r}} = \frac{1}{2}\left(\widehat{\boldsymbol{r}}_{\boldsymbol{a}} + \widehat{\boldsymbol{r}}_{\boldsymbol{b}}\right), \quad (3)$$

and, with our notations, the difference follows the distribution[2]

$$\widehat{\boldsymbol{r}}_{\boldsymbol{a}} - \widehat{\boldsymbol{r}}_{\boldsymbol{b}} \sim \mathcal{N}(\boldsymbol{0}, 4\boldsymbol{\Sigma}). \quad (4)$$

Experimental results on despeckled images obtained with MERLIN [31] support our Gaussian approximation: Figure 3 shows that, after normalizing by the estimated uncertainty $\widehat{\boldsymbol{\sigma}}$ at each pixel, the difference between despeckled images $\widehat{\boldsymbol{r}}_{\boldsymbol{a}}$ and $\widehat{\boldsymbol{r}}_{\boldsymbol{b}}$ is approximately a standard normal.

[2]as $\widehat{\boldsymbol{r}}_{\boldsymbol{a}}$ and $\widehat{\boldsymbol{r}}_{\boldsymbol{b}}$ are independent and identically distributed, by definition (3), if $\text{Cov}(\widehat{\boldsymbol{r}}) = \boldsymbol{\Sigma}$, then $\text{Cov}(\widehat{\boldsymbol{r}}_{\boldsymbol{a}}) = 2\boldsymbol{\Sigma}$.

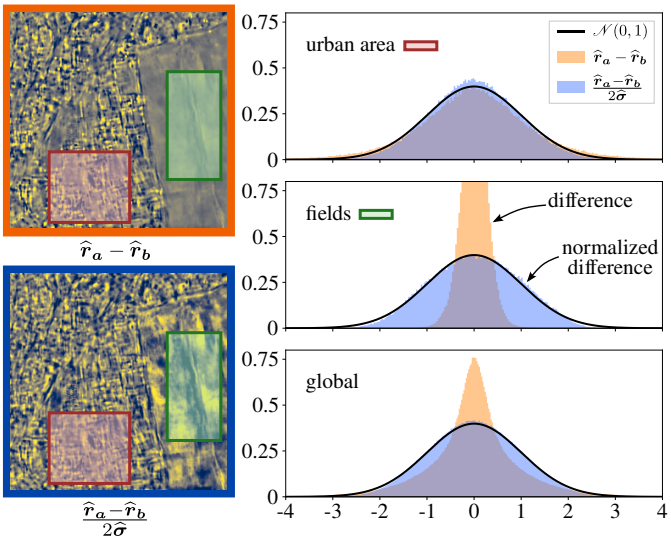


Fig. 3. Illustration of the validity of the Gaussian modeling for despeckled images. The difference $\widehat{\boldsymbol{r}}_{\boldsymbol{a}} - \widehat{\boldsymbol{r}}_{\boldsymbol{b}}$, shown in orange, is not Gaussian. But, after normalization by the estimated standard deviation (bottom row and blue histograms), the distribution is close to standard normal, in urban areas as well as fields or globally on the whole image.

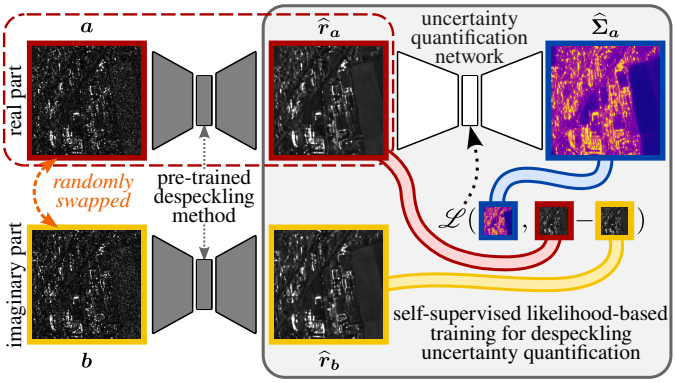


Fig. 4. Training process of the uncertainty quantification network: the decomposition of a complex SAR image into its real and imaginary parts lead to statistically independent reflectivity estimates that can be compared to guide the training of an uncertainty quantification network. See Figure 5 for further details on the architecture of the uncertainty quantification network.

2) Self-supervised estimation of the covariance matrix: We propose to train a neural network[3] $f_{\boldsymbol{\theta}}$ to predict $\boldsymbol{\Sigma}$ from $\widehat{\boldsymbol{r}}_{\boldsymbol{c}}$, with $\boldsymbol{c} \in \{\boldsymbol{a}, \boldsymbol{b}\}$ denoting either the real or imaginary part of the SLC image (Fig. 4). The estimated log-reflectivity image $\widehat{\boldsymbol{r}}_{\boldsymbol{c}}$ contains structural information present in the image (homogeneous areas, borders, textures, point-scatters, *etc.*), from which despeckling reliability can be deduced. Denoting $\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{c}} = f_{\boldsymbol{\theta}}(\boldsymbol{c}, \widehat{\boldsymbol{r}}_{\boldsymbol{c}})$ the prediction of the neural network, we fit the weights $\boldsymbol{\theta}$ to minimize the negative log-likelihood on the training set:

$$\arg\min_{\boldsymbol{\theta}} \sum_{\substack{(\boldsymbol{a},\boldsymbol{b}) \in \text{training set} \\ \boldsymbol{c} \in \{\boldsymbol{a},\boldsymbol{b}\}}} \underbrace{-\log p\left(\widehat{\boldsymbol{r}}_{\boldsymbol{a}} - \widehat{\boldsymbol{r}}_{\boldsymbol{b}} \mid \widehat{\boldsymbol{\Sigma}}_{\boldsymbol{c}}\right)}_{\propto\, \mathscr{L}^{\mathcal{G}}_{\boldsymbol{\theta}}(\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{c}}, \widehat{\boldsymbol{r}}_{\boldsymbol{a}} - \widehat{\boldsymbol{r}}_{\boldsymbol{b}}) + \text{cste}}. \quad (5)$$

[3]an empirical estimator of the covariance $\boldsymbol{\Sigma}$ can be defined neither spatially (covariance is not stationary) nor temporally (due to temporal changes and the lack of samples for large covariance matrices).

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Under our Gaussian model, the loss function writes:

$$\mathscr{L}^{c}_{\boldsymbol{\theta}}(\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{c}}, \widehat{\boldsymbol{r}}_{\boldsymbol{a}} - \widehat{\boldsymbol{r}}_{\boldsymbol{b}}) = \log\det\left(\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{c}}\right) + \frac{1}{4}(\widehat{\boldsymbol{r}}_{\boldsymbol{a}} - \widehat{\boldsymbol{r}}_{\boldsymbol{b}})^{T}\ \widehat{\boldsymbol{\Sigma}}_{\boldsymbol{c}}^{-1}\ (\widehat{\boldsymbol{r}}_{\boldsymbol{a}} - \widehat{\boldsymbol{r}}_{\boldsymbol{b}}). \tag{6}$$

Since the disparity between the two log-reflectivities reflects residual despeckling fluctuations, the difference $\widehat{\boldsymbol{r}}_{\boldsymbol{a}} - \widehat{\boldsymbol{r}}_{\boldsymbol{b}}$ can be compared with the predicted uncertainties to guide training and to learn to estimate the variance (and covariance) of those residual fluctuations. Additionally, since only one despeckled log-reflectivity is provided as input, the neural network can not predict the independent residual fluctuations in the second despeckled log-reflectivity image and can only guess the level of fluctuation. This training technique (used in MERLIN for despeckling) is similar to the Noise2Noise [32] principle. This proposed approach to quantify uncertainties can be applied to most despeckling techniques. On the one hand, statistical despeckling techniques (*i.e.*, not data-driven ones) are often applicable (or adaptable) to arbitrary numbers of looks, in particular to the half-look real and imaginary parts. On the other hand, methods based on deep learning can be trained using only the squared real or the squared imaginary parts, instead of the intensity, to produce the intermediate estimates $\widehat{\boldsymbol{r}}_{\boldsymbol{a}}$ and $\widehat{\boldsymbol{r}}_{\boldsymbol{b}}$. To support this statement, in addition to the results using MERLIN for despeckling, we also provided results using SAR-BM3D [37] in the supplementary materials on GitLab.

In addition, some constraints on the output of $f_{\boldsymbol{\theta}}$ are added to ensure that the predicted matrix is symmetric positive definite. We use the Cholesky decomposition $\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}^{T}$ of the covariance matrix and set our output to correspond to the lower-triangular matrix $\boldsymbol{L}^{-1}$ rather than $\boldsymbol{\Sigma}$. Since $\widehat{\boldsymbol{L}}^{-1}$ must be lower-triangular, we only estimate the coefficients of the lower half of the matrix. Furthermore, we force the diagonal coefficients to be strictly positive by using a softplus activation function. Additionally, our reparameterization with $\widehat{\boldsymbol{L}}^{-1}$ offers a more efficient way to evaluate the loss function (6), avoiding any matrix inversion and leveraging the triangular matrix structure to compute the determinant:

$$\mathscr{L}^{c}_{\boldsymbol{\theta}}\left(\widehat{\boldsymbol{L}}^{-1}_{\boldsymbol{c}}, \widehat{\boldsymbol{r}}_{\boldsymbol{a}} - \widehat{\boldsymbol{r}}_{\boldsymbol{b}}\right) = -\sum_{k=1}^{N^2} \log\left[\widehat{\boldsymbol{L}}^{-1}_{\boldsymbol{c}}\right]_{k,k} + \frac{1}{8}\left\|\widehat{\boldsymbol{L}}^{-1}_{\boldsymbol{c}}(\widehat{\boldsymbol{r}}_{\boldsymbol{a}} - \widehat{\boldsymbol{r}}_{\boldsymbol{b}})\right\|_2^2. \tag{7}$$

3) Neural network architecture: To estimate the covariance matrix that models despeckling uncertainties of a given patch, contextual information beyond the spatial extent of the patch is essential. The input of our neural network covers a large area. To limit the output dimension of the network, we predict despeckling uncertainties for patches in the image *without overlap* (thanks to a downsampling operation). The neural network architecture we propose consists of three parts: a backbone, a neck, and a head (Fig. 5).

- To extract rich features before downsampling, we use a lightweight NAFNet [38] as a backbone, which is a U-shaped architecture that replaces activation functions with internal products of features. Indeed, these internal products are appealing for estimating second-order moments (variance and covariance). This choice has been confirmed by a comparison to U-Net in section III-B.
- The neck is designed to perform downsampling. The network's output is a field of $\widehat{\boldsymbol{L}}^{-1}_{\boldsymbol{c}}$ matrices (one for each patch). Each $\widehat{\boldsymbol{L}}^{-1}_{\boldsymbol{c}}$ is composed of many values that are represented by multiple output channels (for example, $\frac{7^2(7^2+1)}{2} = 1225$ channels for patches of size 7×7). The pixel-shuffle [39] operation (*i.e.*, re-arranging pixels of a 2D patch into a third dimension) achieves a well-suited data-rearrangement of, for example, a 32-channels[4] 256×256 pixels feature map into a 1568-channels 37×37 pixels neck's output. Compared to other alternatives, it performs a downsampling without requiring any learnable weights (in contrast to a strided convolution layer) and without information loss (in contrast to max-pooling or average-pooling). Pixel-shuffle is compared to max-pooling in section III-B.
- Finally, the head is composed of a few convolutions (3×3 and 1×1 kernels to keep it lightweight) to estimate the right number of parameters for $\widehat{\boldsymbol{L}}^{-1}_{\boldsymbol{c}}$.

A particular case of this architecture is when the patch-size is 1×1 (pixel-wise uncertainty): the pixel-shuffle has no effect, and the head projects NAFNet features to an image with only one channel, the variance. In this case, we directly use the loss function (6), which is then computationally cheap.

B. Change detection based on pixels-wise and patch-wise SAR despeckling uncertainties

After despeckling uncertainties have been estimated, they can be leveraged to perform change detection. In the following, we denote the null hypothesis $\mathscr{H}_0$: absence of changes and $\mathscr{H}_1$: presence of a change. We consider two SAR patches of size $N \times N$ acquired at times t_1 and t_2, whose estimated log-reflectivities and despeckling uncertainty covariance matrices[5] are respectively denoted $(\widehat{\boldsymbol{r}}_1, \widehat{\boldsymbol{\Sigma}}_1)$, and $(\widehat{\boldsymbol{r}}_2, \widehat{\boldsymbol{\Sigma}}_2)$. Detecting changes at the scale of patches would degrade the spatial resolution, which would then be limited by the size of the patch. We develop here a method to localize potential changes inside each patch.

The insight for pixel-wise change localization is the following: given the despeckling uncertainties and a desired false alarm rate, at which pixel locations can we consider that the actual log-reflectivity remained constant between the two dates? Additionally, assuming the changes in the patch to be sparse, we propose to solve the following sparse estimation problem (see Fig. 6):

$$\begin{aligned} &(\boldsymbol{r}_1^*, \boldsymbol{r}_2^*) = \underset{\boldsymbol{r}_1, \boldsymbol{r}_2}{\arg\min}\ \|\boldsymbol{r}_1 - \boldsymbol{r}_2\|_1 \\ &\text{s.t.}\quad \|\boldsymbol{r}_1 - \widehat{\boldsymbol{r}}_1\|^2_{\widehat{\boldsymbol{\Sigma}}_1^{-1}} + \|\boldsymbol{r}_2 - \widehat{\boldsymbol{r}}_2\|^2_{\widehat{\boldsymbol{\Sigma}}_2^{-1}} \leq \tau(\eta), \end{aligned} \tag{8}$$

with $\|\boldsymbol{x}\|^2_{\boldsymbol{M}} = \boldsymbol{x}^T\boldsymbol{M}\boldsymbol{x}$ the squared Mahalanobis distance, and $\tau(\eta)$ a threshold that depends on the false alarm rate η. The

[4] $32 = 2^p$ is chosen such that $2^p \cdot N^2$ is close to the number of output channels $\frac{N^2(N^2+1)}{2}$.

[5] covariance matrices are calculated by averaging the covariance matrix estimates from both the real and the imaginary parts.

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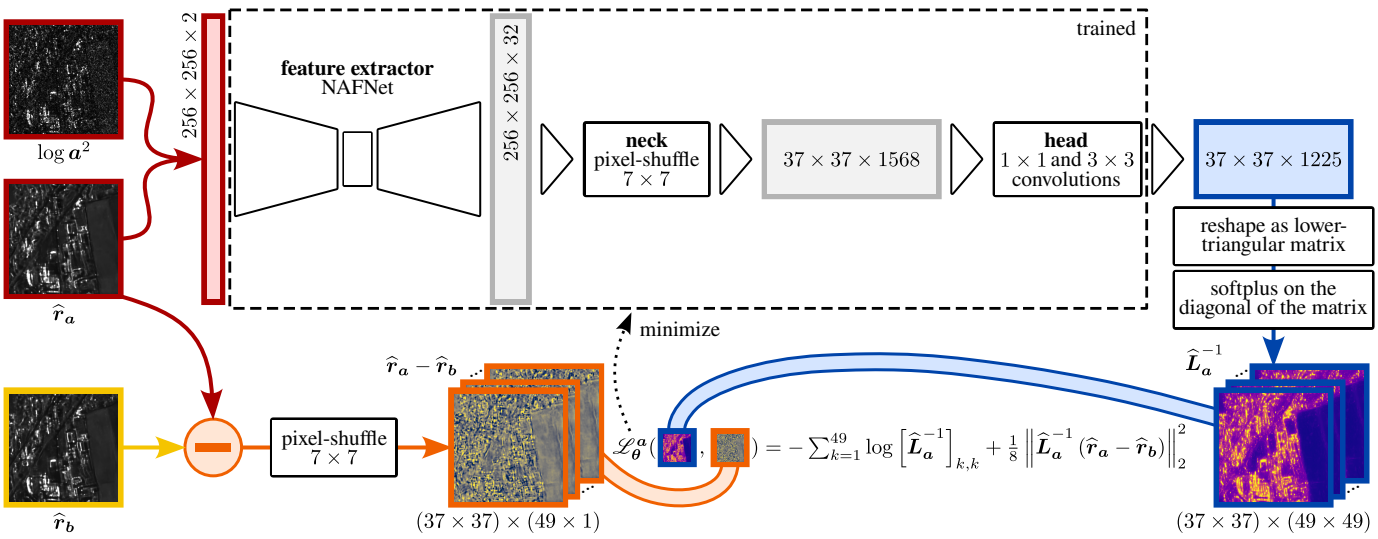


Fig. 5. Illustration of the proposed neural network architecture for an input image of size 256×256 pixels and patches of size 7×7 pixels.

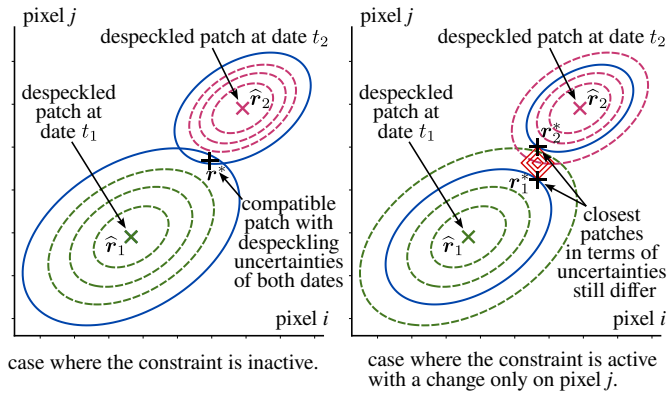


Fig. 6. Geometrical interpretation of the change detection problem for two log-reflectivity patches $\widehat{\boldsymbol{r}}_1$ and $\widehat{\boldsymbol{r}}_2$ composed of two pixels i and j. The blue ellipses delimit the "maximum" allowed distance from the centroids, considering the uncertainties. On the left, the constraint of (8) is not active (intersection of the two ellipses), and no change is detected. On the right, the constraint is active and the minimization of L_1 norm of the change leads to $\boldsymbol{r}_1^*$ and $\boldsymbol{r}_2^*$.

constraint allows the two log-reflectivity patches, $\boldsymbol{r}_1^*$ and $\boldsymbol{r}_2^*$, to differ from the despeckled log-reflectivities by an amount calibrated according to the false alarm rate η.

When the constraint is active, the problem (8) becomes equivalent to the equality problem:

$$\begin{aligned}(\boldsymbol{r}_1^*, \boldsymbol{r}_2^*) &= \underset{\boldsymbol{r}_1, \boldsymbol{r}_2}{\arg\min} \ \|\boldsymbol{r}_1 - \boldsymbol{r}_2\|_1 \\ \text{s.t.} \quad &\|\boldsymbol{r}_1 - \widehat{\boldsymbol{r}}_1\|^2_{\widehat{\boldsymbol{\Sigma}}_1^{-1}} + \|\boldsymbol{r}_2 - \widehat{\boldsymbol{r}}_2\|^2_{\widehat{\boldsymbol{\Sigma}}_2^{-1}} = \tau(\eta).\end{aligned} \tag{9}$$

On the contrary, if the constraint is not active, then the solution is $\boldsymbol{r}_1^* = \boldsymbol{r}_2^*$, *i.e.*, there are no change. Moreover, the constraint is active if and only if $\widehat{\boldsymbol{r}}_1$ is sufficiently different from $\widehat{\boldsymbol{r}}_2$. As detailed in Appendix A, the test deciding of whether or not the constraint is active is:

$$\|\widehat{\boldsymbol{r}}_1 - \widehat{\boldsymbol{r}}_2\|^2_{\left(\widehat{\boldsymbol{\Sigma}}_1 + \widehat{\boldsymbol{\Sigma}}_2\right)^{-1}} \underset{\mathscr{H}_0}{\overset{\mathscr{H}_1}{\gtrless}} \tau(\eta). \tag{10}$$

Thus, we can pre-select patches in which a change has occurred with (10) and, then, localize the changes inside the patch with (9) knowing that the constraint is active. We solve this latter problem thanks to the Lagrangian method and the introduction of a new variable α:

$$\begin{aligned}(\boldsymbol{r}_1^*(\alpha), \boldsymbol{r}_2^*(\alpha)) = \underset{\boldsymbol{r}_1, \boldsymbol{r}_2}{\arg\min} \ &\alpha \|\boldsymbol{r}_1 - \boldsymbol{r}_2\|_1 + \frac{1}{2}\|\boldsymbol{r}_1 - \widehat{\boldsymbol{r}}_1\|^2_{\widehat{\boldsymbol{\Sigma}}_1^{-1}} \\ &+ \frac{1}{2}\|\boldsymbol{r}_2 - \widehat{\boldsymbol{r}}_2\|^2_{\widehat{\boldsymbol{\Sigma}}_2^{-1}} \\ \text{s.t.} \quad &\|\boldsymbol{r}_1^*(\alpha) - \widehat{\boldsymbol{r}}_1\|^2_{\widehat{\boldsymbol{\Sigma}}_1^{-1}} + \|\boldsymbol{r}_2^*(\alpha) - \widehat{\boldsymbol{r}}_2\|^2_{\widehat{\boldsymbol{\Sigma}}_2^{-1}} = \tau(\eta),\end{aligned} \tag{11}$$

which can be tackled by alternating optimization algorithms to find $\boldsymbol{r}_1^*(\alpha)$ and $\boldsymbol{r}_2^*(\alpha)$ for a fixed value of α, and root-finding steps to progressively determine the correct value of α based on successive solutions of $\boldsymbol{r}_1^*(\alpha)$ and $\boldsymbol{r}_2^*(\alpha)$. Indeed, thanks to the first line of (11), $\|\boldsymbol{r}_1^*(\alpha) - \widehat{\boldsymbol{r}}_1\|^2_{\widehat{\boldsymbol{\Sigma}}_1^{-1}} + \|\boldsymbol{r}_2^*(\alpha) - \widehat{\boldsymbol{r}}_2\|^2_{\widehat{\boldsymbol{\Sigma}}_2^{-1}}$ is increasing as a function of α. The optimization process is described in the algorithm 1. Further details on the chosen optimization process and the false alarm calibration to select $\tau(\eta)$ are given in Appendices B and C.

Remark: in the case of patches of size 1×1 pixels, the solution at each pixel is trivial and is the Generalized Likelihood Ratio Test (GLRT):

$$\text{GLRT} : \frac{(\widehat{r}_1 - \widehat{r}_2)^2}{\widehat{\sigma}_1^2 + \widehat{\sigma}_2^2} \underset{\mathscr{H}_0}{\overset{\mathscr{H}_1}{\gtrless}} \gamma(\eta), \tag{12}$$

with $\widehat{\sigma}_i^2$ the estimated variance of the estimated log-reflectivity $\widehat{r}_i$ at each pixel, and where $\gamma(\eta)$ can be set thanks to the $\chi^2(1)$ cumulative distribution function (no calibration required).

III. Experiments and Analysis

The proposed SAR-DUCK is tested on both TerraSAR-X Stripmap and on Sentinel-1 Interferometric Wide swath images.

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Algorithm 1 Algorithm used to solve (11).

Inputs $\tau(\eta)$, # Threshold for the false alarm rate η
$\widehat{\boldsymbol{r}}_1, \widehat{\boldsymbol{r}}_2$, # Patches of log-reflectivities
$\widehat{\boldsymbol{\Sigma}}_1^{-1}, \widehat{\boldsymbol{\Sigma}}_2^{-1}$ # Estimated uncertainties

Initialize α;
$\delta(\alpha) \leftarrow +\infty$;

while $|\delta(\alpha)| > 10^{-6}$ **do**
  # Optimization of the first line of (11):
  $\boldsymbol{r}_1^*(\alpha), \boldsymbol{r}_2^*(\alpha) \leftarrow$ optimization$(\alpha, \widehat{\boldsymbol{r}}_1, \widehat{\boldsymbol{r}}_2)$;

  # Constraint error, second line of (11):
  $\delta(\alpha) \leftarrow \|\boldsymbol{r}_1^*(\alpha) - \widehat{\boldsymbol{r}}_1\|^2_{\widehat{\boldsymbol{\Sigma}}_1^{-1}} + \|\boldsymbol{r}_2^*(\alpha) - \widehat{\boldsymbol{r}}_2\|^2_{\widehat{\boldsymbol{\Sigma}}_2^{-1}} - \tau(\eta)$;
  $\alpha \leftarrow$ iteration of root-finding $(\alpha, \delta(\alpha))$;
end while

Returns $\boldsymbol{r}_1^*(\alpha), \boldsymbol{r}_2^*(\alpha)$

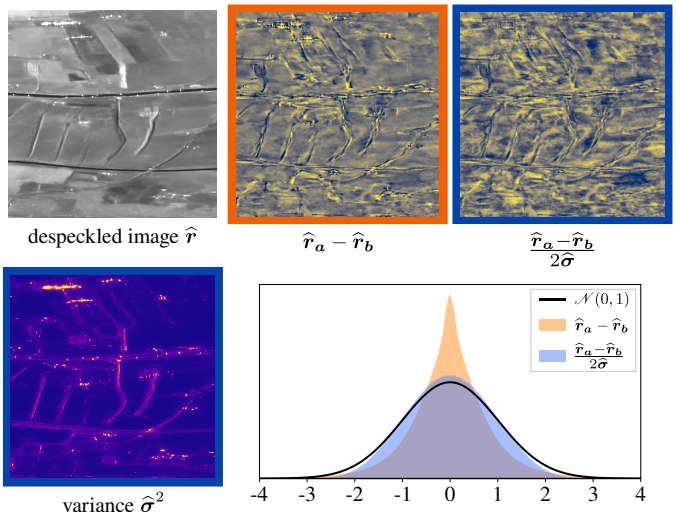


Fig. 7. Illustration of the quality of the pixel-wise variance estimate on Sentinel-1 images. On top, a despeckled image (left) and the difference between the estimates computed only from the real or the imaginary part (center).This difference is then normalized by the estimated pixel-wise variance (right). The histogram of unnormalized and normalized differences (on a larger area) is shown on the bottom row, with the estimated variance.

A. Datasets

We use the weights of a neural network trained to despeckle TerraSAR-X Stripmap images, available on the *deepdespeckling* repository[6]. To train our network for despeckling uncertainty quantification, two regions of interest of size $1,024 \times 1,024$ pixels in addition to 22 images of approximately $30,000 \times 16,000$ pixels, were used to generate $266,386$ training patches and $7,522$ validation patches of size 256×256. Temporal pairs from the two predefined regions of interest (these specific acquisitions are neither in the training nor in the validation dataset) are used to qualitatively and quantitatively evaluate the localization of changes on TerraSAR-X images. The ground truths of changes in these pairs were manually labeled using both optical images, for better scene understanding, and time series, to better distinguish speckle from actual temporal changes.

To process Sentinel-1 images, SNAP is first used to prepare the images by selecting the first swath[7], perform deramping (required to obtain independent speckle in the real and the imaginary components of an image acquired in TOPSAR mode), radiometric calibration σ_0, de-bursting, and coregistration of pairs of images. These pre-processing steps are applied to 25 images of approximately $12,000 \times 20,000$ pixels to generate $85,947$ training patches and $4,512$ validation patches of size 256×256 pixels. Both the MERLIN despeckling network and the uncertainty quantification network are trained on this dataset. Finally, four regions of interest are particularly studied for the Sentinel-1 detection and localization of changes across various areas of the Amazonian forest in Brazil. As both despeckling and uncertainty quantification networks are trained in a self-supervised manner and not fed with image pairs, they can be fine-tuned on test images with no risk of overfitting change detection. Consequently, the first-date images of the tested pairs are included in the training or validation dataset.

Most traditional SAR change detection datasets (Ottawa, Sulzberger, Yellow-river, San-Francisco, Farmland, Bern, *etc.*) are multi-looked and quantized to 256 gray levels, raising further questions about generalization capabilities [40]. In contrast, we evaluate the methods on single-look image pairs, and we preserve the original radiometric resolution of 32 bits, as typically provided by space agencies.

B. Analysis of the quantification of despeckling uncertainties

First, we test the accuracy of the pixel-wise quantification of despeckling uncertainties. To do so, we normalize residual fluctuations of despeckling in absence of changes. Figure 7 presents the histogram of the normalized difference of log-reflectivity computed on a large $4,096 \times 4,096$ pixels image. This figure illustrates that the normalization stabilizes the residual fluctuations as the histogram reflects a good fit with a Gaussian model.

Moreover, we examine the relevance of the covariance estimates by whitening the differences by the estimated uncertainties. We show in the Figure 8 the whitening of the same area for a patch-size from 1×1 to 7×7. As the patch size of the whitening increases, both the visible structures in the normalized difference progressively disappear, and the range of the autocorrelations becomes progressively smaller. This reflects an improvement of the covariance estimates. This refinement of the Gaussian model is confirmed by the validation loss during the training on TerraSAR-X and Sentinel-1 images (Tab. I). Note that all trainings were conducted with the same dataset and the same number of epochs. In contrast to the gradual increase in size from 1×1 up to 7×7, patches of size 9×9 pixels offer only a slight improvement over 7×7 patches. This can be explained by the fact that the

[6]https://github.com/hi-paris/deepdespeckling/ - with the support of Hi! PARIS and ANR/France 2030 program (ANR-23-IACL-0005)

[7]training a neural network dedicated to each swath may be beneficial as their resolutions differ.

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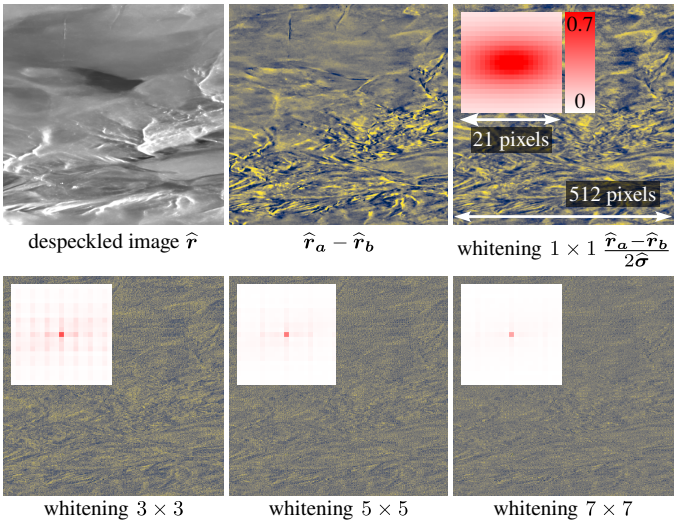


Fig. 8. Effect of the whitening $\widehat{\boldsymbol{L}}^{-1}$ for a patch-size from 1×1 to 7×7 pixels. The thumbnails in the upper-left corner of each image indicate the empirical autocorrelation of the whitened differences. As the whitening improves, the autocorrelation tends to a Dirac.

TABLE I
NEGATIVE LOG-LIKELIHOOD OF THE GAUSSIAN MODELS, WITH THE COVARIANCES PREDICTED BY OUR NETWORK, EVALUATED ON TERRASAR-X AND SENTINEL-1 VALIDATION SETS, DEPENDING ON THE PATCH SIZE (THE LOWER THE BETTER).

Sensor	1×1	3×3	5×5	7×7	$\mathbf{9 \times 9}$
TerraSAR-X	-0.037	-1.394	-1.748	-1.915	$\mathbf{-1.993}$
Sentinel-1	-0.274	-1.595	-1.958	-2.138	$\mathbf{-2.223}$

training convergence is slower: a larger number of epochs would be required to further improve the network that predicts despeckling uncertainties with bigger patches. Patch-sizes 5×5 and 7×7 offer a good compromise between model precision and learning complexity. Finally, Table II compares several architectures on Sentinel-1 images. Slightly lower negative log-likelihood values indicate that the NAFNet architecture is better adapted for this task than U-Net, even with fewer weights. It also shows that, compared to max-pooling, the pixel-shuffle operation significantly improves performance, for a very close number of network weights.

C. Analysis of the proposed change detection

The proposed change detection method is evaluated qualitatively and quantitatively on two TerraSAR-X pairs, and only qualitatively on several Sentinel-1 Amazonian pairs with more subtle radiometric changes. SAR-DUCK is compared to four unsupervised SAR change-detection methods of reference from the state-of-the-art. On the one hand, PCA k-means [8] is a traditional method that combines PCA on patches with k-means on the projected features, and SRF [41] is a non deep-learning-based recent multimodal change detection method. On the other hand, FFITN [18], SAFNet [17], and WBANet [42] are recent change detection methods that train a neural network on pseudo-labels, and for which the code is available online. Note that these methods have not originally been tested on single-look SAR images, which are noisier than multi-looked images. Thus, we have applied those methods both to single-look images and to images despeckled by MERLIN [31], and we have also searched for parameter settings that led to the best results in each case. To illustrate the importance of despeckling uncertainties for change detection, we also compare our method to a simpler baseline: MERLIN despeckling followed by thresholding at 5% false alarms (threshold selected to achieve good-looking results). To round out the experiments, three other methods (INLPG [43], LANTNet [44], and CAMixer [45]) are compared quantitatively in the results tables. Also note that the number of training epochs for FFITN and the threshold value for INLPG are not determined in an unsupervised manner in their implementation. For TerraSAR-X data, using the ground truth, we set those values to maximize the κ coefficient (defined in the following paragraph, Eq. (18)), as originally set for FFITN. Regarding the lack of ground truth, we set those values to achieve the best visual results with Sentinel-1 data.

TABLE II
COMPARISON OF SEVERAL NETWORK ARCHITECTURES FOR UNCERTAINTY PREDICTION: NEGATIVE LOG-LIKELIHOOD VALUES COMPUTED ON THE SENTINEL-1 VALIDATION SET, FOR 5×5 PATCHES.

Body	U-Net		U-Net heavy		**NAFNet**	
Neck*	M.P.	P.S.	M.P.	P.S.	M.P.	**P.S.**
Weights**	1.3M	1.3M	2.6M	2.6M	2.2M	**2.2M**
Loss	-1.841	-1.941	-1.848	-1.933	-1.866	$\mathbf{-1.958}$

* "M.P." and "P.S." stands for max-pooling and pixel-shuffle.
** "M" indicates that the amounts of weights are in millions.

Figure 9 shows the performance of the compared change detection methods on two TerraSAR-X pairs. We clearly see that despeckling greatly facilitates change detection, especially to detect low-radiometry changes in the fields. Additionally, thanks to the quantification of the despeckling uncertainties, our method (calibrated to a false alarm rate $\eta = 10^{-5}$) raises much fewer false alarms in highly textured areas such as the city than reference methods. Finally, accounting for the spatial correlation of the despeckling uncertainties appears to improve the detection of highly structured changes, such as buildings, as seen in the middle-left of the Saint-Gervais area. Quantitative performances provided in the Tables III and IV confirms those observations for six metrics: (i) the False Alarm Rate (FAR); (ii) the Overall Error (OE); (iii) the precision (Pre), which is the rate of true detected changes among all detections; (iv) the recall (Rec), which is the rate of true detected changes among all true changes; (v) the F_1 score that measures how the precision and the recall are balanced, which is useful for unbalanced classification problems such as change detection; (vi) finally, the κ coefficient, which broadly measures how the change detection is consistent with the ground truth. With TP, TN, FP, and FN denoting, respectively, the true positive, true negative, false positive, and false negative, we have:

$$\mathrm{FAR} = \frac{\mathrm{FP}}{\mathrm{TN} + \mathrm{FP}}, \tag{13}$$

$$\mathrm{OE} = \frac{\mathrm{FP} + \mathrm{FN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}, \tag{14}$$

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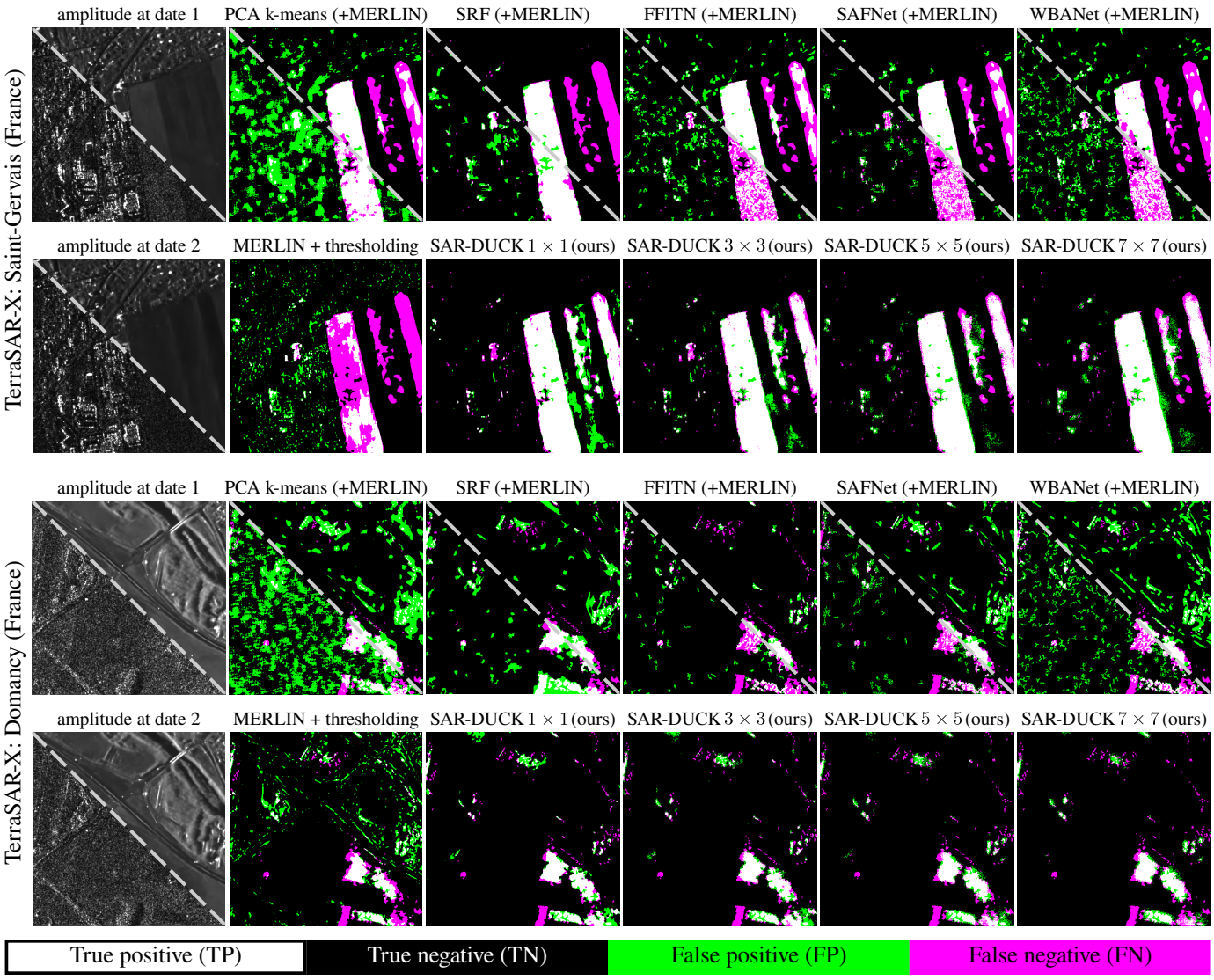


Fig. 9. Qualitative radiometric change detection results on real TerraSAR-X data: Saint-Gervais and Domancy (France) regions of interest. For both studied areas, the left images show the amplitude (bottom left) and the square root of the reflectivity estimated by MERLIN [31] (top right). The other images in the first row for each dataset are results provided by reference unsupervised SAR change detection methods. Even if those methods do not originally use a despeckling technique, we show the benefit of applying a despeckling step as pre-processing (MERLIN). From left to right: PCA k-means [8], SRF [41], FFITN [18], SAFNet [17] and WBANet [42]. Finally, the other results on the second row are the ones of the pipeline MERLIN + thresholding, and of SAR-DUCK (ours) with different patch-sizes, from 1×1 on the left to 7×7 on the right.

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \tag{15}$$

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \tag{16}$$

$$\text{F}_1 = 2\frac{\text{Pre} \cdot \text{Rec}}{\text{Pre} + \text{Rec}}, \tag{17}$$

$$\kappa = \frac{1 - \text{OE} - \text{Pe}}{1 - \text{Pe}}, \tag{18}$$

with

$$\text{Pe} = \frac{(\text{TP} + \text{FP})(\text{TP} + \text{FN}) + (\text{TN} + \text{FP})(\text{TN} + \text{FN}))}{(\text{TP} + \text{TN} + \text{FP} + \text{FN})^2}. \tag{19}$$

Note that a high precision with a low recall reflects a lack of detection, while a high recall with a low precision indicates a tendency to over-detect changes. The table also underlines that the method is very competitive even for very small patches. Using large patches seems detrimental, due to the increased difficulty to estimate the covariance matrices modeling despeckling uncertainties, the border effects inside each patch (further discussed in Appendix B), and the lack of spatial regularization of the L_1 norm. Using small patches of size 3×3 or 5×5 pixels appears to capture sufficiently complex correlations in despeckling uncertainties for change detection (borders or orientation of objects, *etc.*).

Figure 10 compares qualitatively the proposed method with the reference methods on Sentinel-1 pairs of images. With this different satellite, operating at a coarser spatial resolution, different acquisition mode (TOPSAR), and frequency band (C-band *vs.* X-band), we reach the similar conclusions as for TerraSAR-X images: both despeckling and uncertainty quantification are very beneficial for, respectively, detecting

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TABLE III
RADIOMETRIC CHANGE DETECTION PERFORMANCES ON TERRASAR-X DATA: SAINT-GERVAIS PAIR (FRANCE).

Methods	FAR (%) ↓	OE (%) ↓	Pre (%) ↑	Rec (%) ↑	F_1 (%) ↑	κ (%) ↑	evaluation time*
PCA k-means [8]	27.84	27.61	36.08	73.44	48.39	32.41	3.60s
INLPG [43]	7.369	16.60	53.89	40.25	46.09	36.52	34.5s
SRF [41]	3.975	9.846	77.15	62.72	69.19	63.41	2.41s
LANTNet [44]	8.953	16.43	53.76	48.64	51.07	41.23	143s
CAMixer [45]	6.619	16.17	55.91	39.22	46.10	36.94	40.9s
FFITN [18]	6.759	15.91	56.68	41.34	47.81	38.69	894s
SAFNet [17]	3.124	13.15	73.25	39.97	51.72	44.86	190s
WBANet [42]	10.59	18.34	47.88	45.45	46.63	35.57	97.7s
MERLIN + thresholding	4.958	16.77	54.75	28.03	37.07	28.54	**0.0332s**
MERLIN + PCA k-means [8]	15.19	17.81	49.62	69.91	58.04	47.15	2.37s
MERLIN + INLPG [43]	6.559	14.10	62.30	50.65	55.87	47.58	32.9s
MERLIN + SRF [41]	**2.271**	8.748	85.18	60.99	71.08	66.09	1.82s
MERLIN + LANTNet [44]	8.631	9.524	68.15	86.30	76.16	70.31	153s
MERLIN + CAMixer [45]	7.143	10.83	68.30	71.93	70.07	63.46	42.8s
MERLIN + FFITN [18]	6.001	8.647	73.80	78.99	76.31	71.03	733s
MERLIN + SAFNet [17]	4.275	7.969	78.92	74.77	76.78	71.98	211s
MERLIN + WBANet [42]	6.350	11.90	67.68	62.15	64.80	57.66	101s
SAR-DUCK 1×1 (12) (ours)	4.302	5.365	81.69	**89.67**	85.49	82.21	0.0726s
SAR-DUCK 3×3 (8) (ours)	2.821	**4.649**	**86.81**	86.81	**86.81**	**83.99**	2630s
SAR-DUCK 5×5 (8) (ours)	2.802	4.949	86.65	85.02	85.83	82.83	3030s
SAR-DUCK 7×7 (8) (ours)	3.365	5.356	84.44	85.34	84.89	81.63	3820s

* to process a pair of 384×384 pixels images on a NVIDIA RTX 2000 Ada Laptop GPU. PCA k-mean, INLPG, and SRF implementations run on the CPU.

TABLE IV
RADIOMETRIC CHANGE DETECTION PERFORMANCES ON TERRASAR-X DATA: DOMANCY PAIR (FRANCE).

Methods	FAR (%) ↓	OE (%) ↓	Pre (%) ↑	Rec (%) ↑	F_1 (%) ↑	κ (%) ↑	evaluation time*
PCA k-means [8]	35.62	35.09	11.84	73.00	20.38	10.95	2.82s
INLPG [43]	2.088	5.755	54.59	38.30	45.02	42.08	33.8s
SRF [41]	8.452	9.559	36.32	73.55	48.63	44.02	1.94s
LANTNet [44]	13.01	14.77	22.75	58.47	32.76	26.22	146s
CAMixer [45]	8.061	10.78	27.96	47.74	35.27	29.82	41.5s
FFITN [18]	5.246	8.842	31.20	36.30	33.55	28.85	931s
SAFNet [17]	5.403	9.009	30.38	35.96	32.93	28.14	204s
WBANet [42]	12.13	14.31	22.09	52.46	31.09	24.56	111s
MERLIN + thresholding	5.941	8.256	38.37	56.43	45.68	41.38	**0.0332s**
MERLIN + PCA k-means [8]	12.11	12.71	29.74	78.21	43.09	37.52	2.38s
MERLIN + INLPG [43]	2.935	5.471	55.50	55.83	55.66	52.75	32.2s
MERLIN + SRF [41]	4.969	6.961	45.25	62.65	52.54	48.89	1.56s
MERLIN + LANTNet [44]	14.56	14.66	27.40	**83.84**	41.30	35.30	156s
MERLIN + CAMixer [45]	12.65	13.27	28.59	77.29	41.74	36.00	47.4s
MERLIN + FFITN [18]	2.005	4.422	65.74	58.71	62.03	59.69	772s
MERLIN + SAFNet [17]	6.100	7.650	42.46	68.69	52.48	48.57	224s
MERLIN + WBANet [42]	9.235	10.41	33.71	71.64	45.84	40.90	104s
SAR-DUCK 1×1 (12) (ours)	2.041	**4.013**	**67.91**	65.90	**66.89**	**64.75**	0.0726s
SAR-DUCK 3×3 (8) (ours)	2.028	4.178	67.07	63.03	64.99	62.77	1260s
SAR-DUCK 5×5 (8) (ours)	1.929	4.463	65.90	56.88	61.06	58.70	1590s
SAR-DUCK 7×7 (8) (ours)	**1.663**	4.414	67.88	53.61	59.90	57.60	2040s

* to process a pair of 384×384 pixels images on a NVIDIA RTX 2000 Ada Laptop GPU. PCA k-mean, INLPG, and SRF implementations run on the CPU.

low-radiometry changes[8] and reducing the number of false alarms in critical environments. However, because we tested the method on forested areas with lower spatial variations of the despeckling uncertainties (*i.e.*, more homogeneous images), the improvement induced by uncertainty quantification is not as significant as observed in the TerraSAR-X images. Change heatmaps and corresponding detected changes are shown in Figure 11 using pixel-wise despeckling uncertainty quantification (the variance at each pixel). From a computational point of view, the pixel-wise method is well-suited to large volumes of data (large images or long time series). Indeed, the optimization of the change detection problem is known in close-form (Eq. (12)). As a result, this method is very fast in inference: less than 1s/Mpixel with a NVIDIA A40 GPU.

An example of a false alarm heatmap and of the distribution of the number of false alarms is presented in Figure 12.

[8]as the task is different and changes are more subtle, the false alarm probability was set to $\eta = 10^{-4}$ for these Sentinel-1 images.

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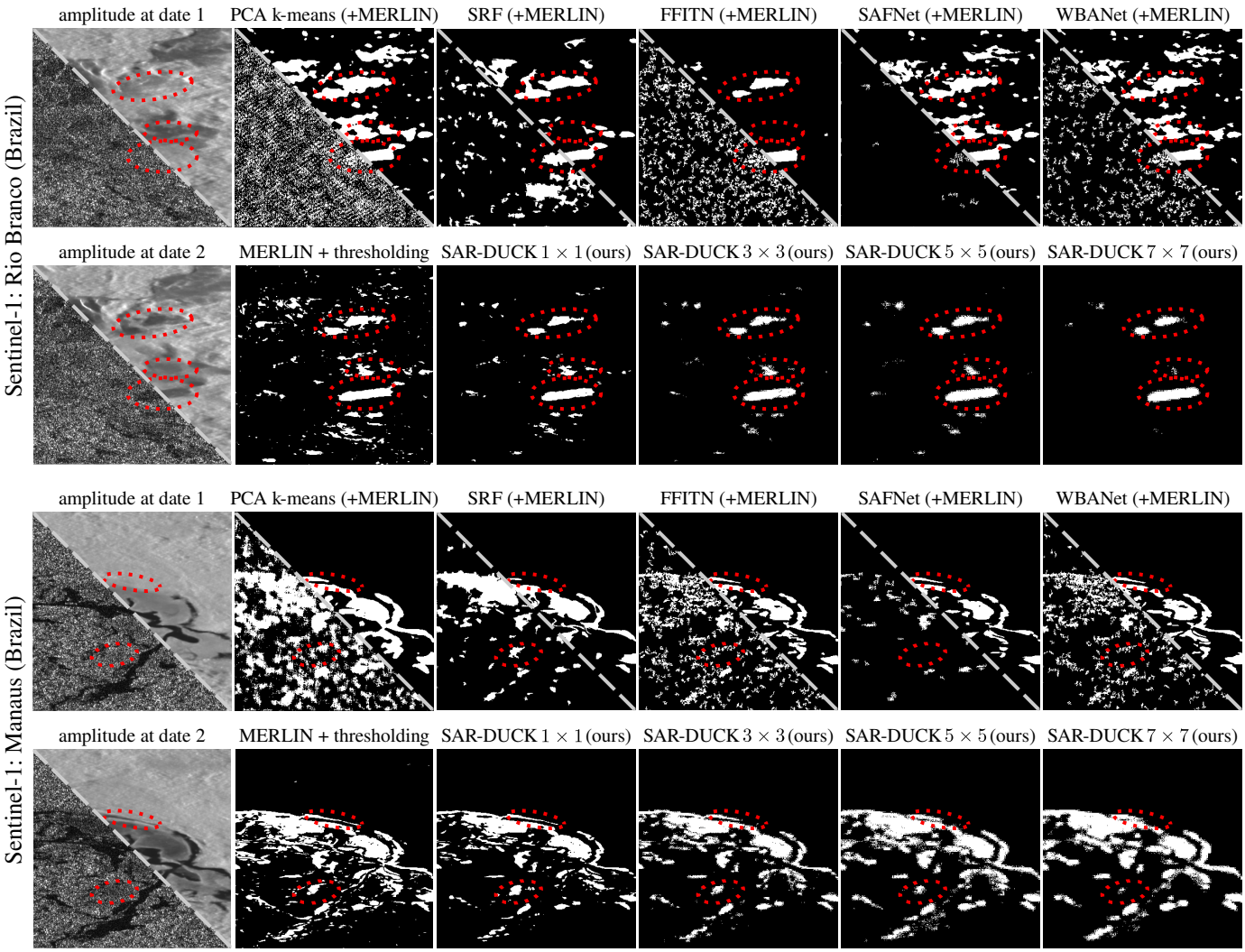


Fig. 10. Qualitative radiometric change detection results on Sentinel-1 region of interest near Rio Branco and Manaus (Brazil). Date 1 of Rio Branco and of Manaus are respectively part of the validation and of the training set. The left images are the ones we research changes in (noisy at bottom left and despeckled at top right). The other images on the first lines are results provided by reference unsupervised SAR change detection methods. Finally, the other images on the second row are results provided by MERLIN + thresholding pipeline and the proposed SAR-DUCK with different patch-sizes from 1×1 on the left to 7×7 on the right.

The heatmap shows the spatially dispersed distribution of false alarms across several regions, including forests, fields, a city, and mountains. The boxplots on the right attest to good consistency in the false alarm rate across a sliding window, around the theoretical one η. Those two observations show that the proposed method is CFAR. Given one date and a fixed false alarm probability η, the close-form proposed pixel-wise change detection provides a way to compute the theoretical probability of detecting a change with a given reflectivity contrast c (*i.e.*, $\exp(\widehat{\boldsymbol{r}}_2) = c \cdot \exp(\widehat{\boldsymbol{r}}_1)$). Alternatively, we can compute the minimal contrast that can theoretically be detected with a probability at least p_s. Figure 13 shows an example of such sensitivity maps on a Sentinel-1 image.

We compare in Figure 14 (rows 2 and 4) coarse change detection at the scale of patches, which loses spatial resolution, to the proposed pixel-level optimization (8). Instead of performing the whole proposed change detection method, for a given pixel, coarse change detection is obtained by the equation (10) (selecting patches where we localize changes with (11)) applied to the patch centered on the considered pixel. This figure shows how the change localization problem allows the exclusion of some unchanged pixels that were detected by the coarse method. Additionally in Figure 14 (rows 3 and 4), we compare the results obtained by the proposed optimization problem (8), or by an alternative optimization problem where the sparsity assumption is relaxed by replacing the L_1 norm by a L_2 norm:

$$\begin{aligned} (\boldsymbol{r}_1^*, \boldsymbol{r}_2^*) &= \underset{\boldsymbol{r}_1, \boldsymbol{r}_2}{\arg\min} \ \|\boldsymbol{r}_1 - \boldsymbol{r}_2\|_2^2 \\ \text{s.t.} \quad & \|\boldsymbol{r}_1 - \widehat{\boldsymbol{r}}_1\|^2_{\widehat{\boldsymbol{\Sigma}}_1^{-1}} + \|\boldsymbol{r}_2 - \widehat{\boldsymbol{r}}_2\|^2_{\widehat{\boldsymbol{\Sigma}}_2^{-1}} \leq \tau(\eta), \end{aligned} \tag{20}$$

with the same value $\tau(\eta)$ as used in (10) (even if it does not correspond to the same false alarm rate) for an easier comparison. We can see in the figure that, if no sparsity assumption is made, all the pixels of a patch that is detected as changed are systematically detected as changed. Thus, it is as coarse as a patch-level change detection.

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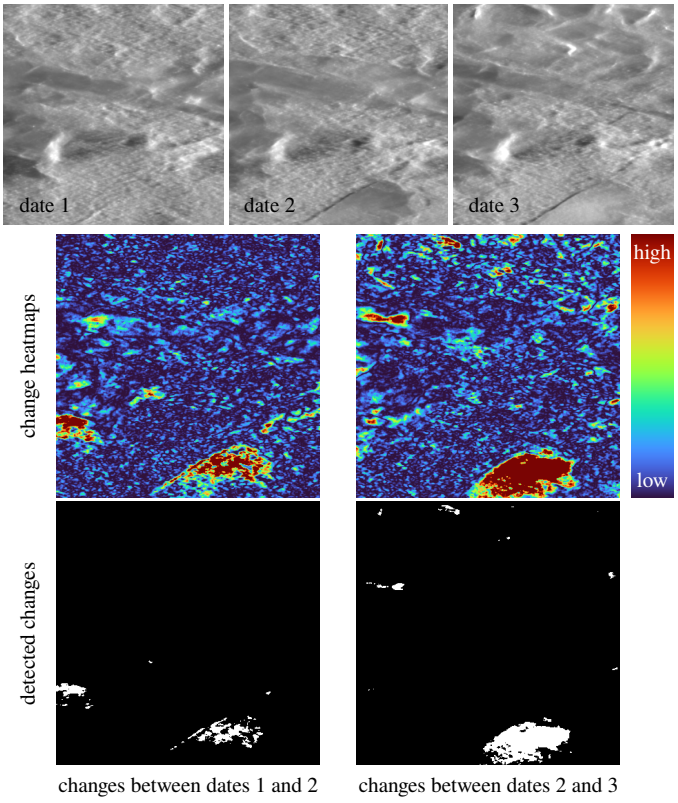


Fig. 11. Proposed pixel-wise change detection method on a Sentinel-1 images near Altamira (Brazil). The images on the first row are despeckled, and date 1 is part of the training set.

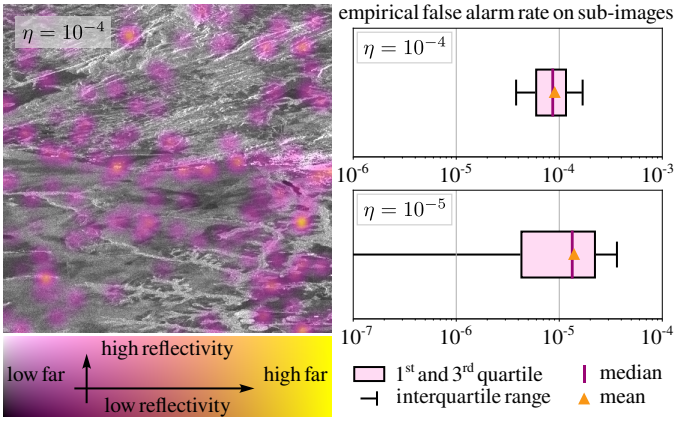


Fig. 12. Study of the CFAR property on a Sentinel-1 image near Brasilia (Brazil) that contains several landscapes (original size of $\sim 4000 \times 4000$ pixels). A no-change pair can be formed from the real and imaginary parts on the same date. A first date is used to calibrate $\tau(\eta)$, our experiment is run on a second date. On the left, the downsampled amplitude is overlaid with the spatial heatmap of false alarms for qualitative evaluation (gray levels correspond to no false alarms). On the right, boxplots depict the empirical false alarm rate in a 1024×1024 pixels sliding window for (top) $\eta = 10^{-4}$ and (bottom) $\eta = 10^{-5}$ (false alarm rate 10^{-5} corresponds to 10.5 pixels).

IV. Conclusion

In this article, we proposed a despeckling uncertainty quantification that accounts for spatial correlations in the uncertainties, and an unsupervised change detection technique between pairs of SAR images that is robust to speckle and reaches a controlled and constant false alarm rate.

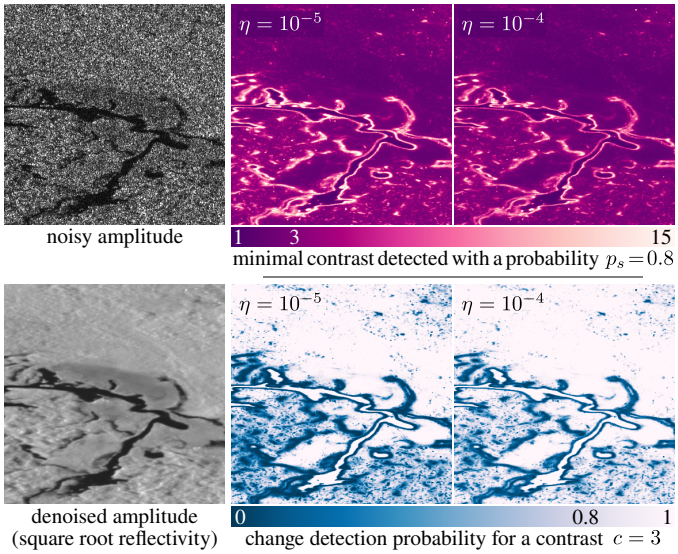


Fig. 13. Sensitivity maps computed for the Sentinel-1 image of Manaus, Brazil, shown on the left column: (top-right) the minimum reflectivity contrast (*i.e.*, not in log-reflectivity scale) detected with probability $p_s = 0.8$, and (bottom-right) the probability to detect a change of contrast $c = 3$ (in reflectivity scale) are shown. The middle column corresponds to a false alarm probability $\eta = 10^{-5}$, and the right column to $\eta = 10^{-4}$.

We showed that the proposed modeling of despeckling errors as a Gaussian distribution is accurate, and that we can quantify despeckling uncertainties by training a neural network in a self-supervised manner similar to Noise2Noise. Moreover, the estimation of the spatial correlations of those uncertainties is also demonstrated to be efficient as shown in whitening experiments.

The model of these despeckling uncertainties is included in our unsupervised change detection method SAR-DUCK. It achieves state-of-the-art results for radiometric change detection on both TerraSAR-X and Sentinel-1 images. Using spatial correlations of the despeckling uncertainties improves the detection of highly structured changes such as buildings, even with patches as small as 3×3 pixels. The pixel-wise change detection method is very fast and well-suited to large images or long time series. Future extensions of the method might include adding intra- and inter-patch spatial regularization to further improve change detection in structured scenes.

Appendix A
Condition for the constraint to be active in the problem of localization of changes

The localization of changes within a patch has been formulated as the following constrained optimization problem:

$$\begin{aligned} (\boldsymbol{r}_1^*, \boldsymbol{r}_2^*) &= \underset{\boldsymbol{r}_1, \boldsymbol{r}_2}{\arg\min} \ \|\boldsymbol{r}_1 - \boldsymbol{r}_2\|_1 \\ \text{s.t.} \quad & \|\boldsymbol{r}_1 - \widehat{\boldsymbol{r}}_1\|^2_{\widehat{\boldsymbol{\Sigma}}_1^{-1}} + \|\boldsymbol{r}_2 - \widehat{\boldsymbol{r}}_2\|^2_{\widehat{\boldsymbol{\Sigma}}_2^{-1}} \leq \tau(\eta). \end{aligned} \tag{21}$$

When the constraint is not active, the solution $(\boldsymbol{r}_1^*, \boldsymbol{r}_2^*)$ is such that $\boldsymbol{r}_1^* = \boldsymbol{r}_2^* = \boldsymbol{r}^*$. Among all vectors $\boldsymbol{r}^*$, the one that satisfies the inequality constraint with the largest margin is the vector such that

$$\boldsymbol{\nabla}_{\boldsymbol{r}^*} \left[\|\boldsymbol{r}^* - \widehat{\boldsymbol{r}}_1\|^2_{\widehat{\boldsymbol{\Sigma}}_1^{-1}} + \|\boldsymbol{r}^* - \widehat{\boldsymbol{r}}_2\|^2_{\widehat{\boldsymbol{\Sigma}}_2^{-1}} \right] = \boldsymbol{0}, \tag{22}$$

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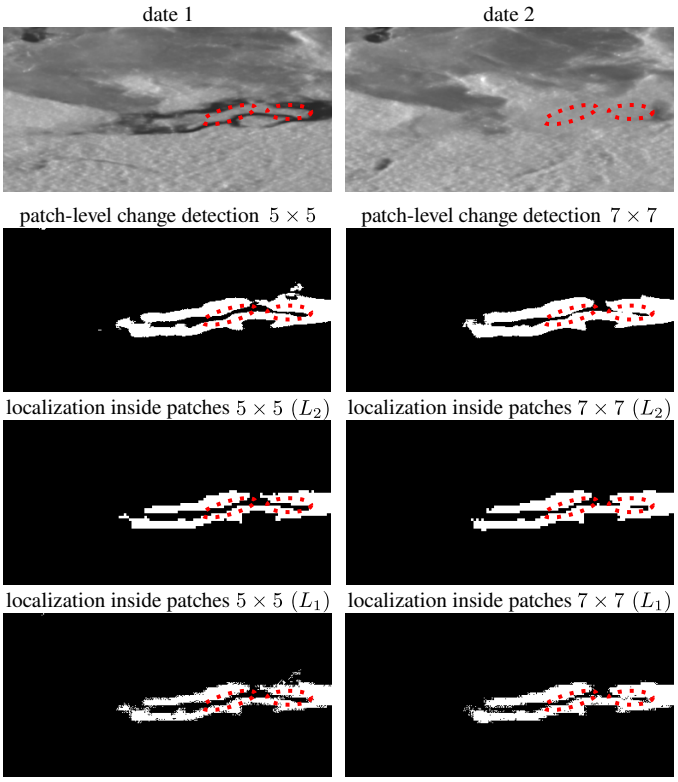


Fig. 14. Comparison of coarse and fine change detection on a pair of Sentinel-1 images acquired near Morais Almeida, Brazil. From top to bottom, despeckled pair of Sentinel-1 images, coarse change detection at the patch-level (10), coarse change detection without the sparsity assumption (20), and proposed pixel-level localization from the equation (8) (Eq. (10) + Eq.(11)).

i.e., the vector:

$$\boldsymbol{r}^* = \left(\widehat{\boldsymbol{\Sigma}}_1^{-1} + \widehat{\boldsymbol{\Sigma}}_2^{-1}\right)^{-1} \left(\widehat{\boldsymbol{\Sigma}}_1^{-1}\widehat{\boldsymbol{r}}_1 + \widehat{\boldsymbol{\Sigma}}_2^{-1}\widehat{\boldsymbol{r}}_2\right). \quad (23)$$

Substituting this expression of $\boldsymbol{r}^*$ in equation (21) leads to

$$\left\| \left(\widehat{\boldsymbol{\Sigma}}_1^{-1} + \widehat{\boldsymbol{\Sigma}}_2^{-1}\right)^{-1} \left(\widehat{\boldsymbol{\Sigma}}_1^{-1}\widehat{\boldsymbol{r}}_1 + \widehat{\boldsymbol{\Sigma}}_2^{-1}\widehat{\boldsymbol{r}}_2\right) - \widehat{\boldsymbol{r}}_1 \right\|^2_{\widehat{\boldsymbol{\Sigma}}_1^{-1}} + \left\| \left(\widehat{\boldsymbol{\Sigma}}_1^{-1} + \widehat{\boldsymbol{\Sigma}}_2^{-1}\right)^{-1} \left(\widehat{\boldsymbol{\Sigma}}_1^{-1}\widehat{\boldsymbol{r}}_1 + \widehat{\boldsymbol{\Sigma}}_2^{-1}\widehat{\boldsymbol{r}}_2\right) - \widehat{\boldsymbol{r}}_2 \right\|^2_{\widehat{\boldsymbol{\Sigma}}_2^{-1}} < \tau(\eta), \quad (24)$$

which leads, after simplifications, to:

$$\|\widehat{\boldsymbol{r}}_1 - \widehat{\boldsymbol{r}}_2\|^2_{\widehat{\boldsymbol{\Sigma}}_1^{-1}\left(\widehat{\boldsymbol{\Sigma}}_1^{-1}+\widehat{\boldsymbol{\Sigma}}_2^{-1}\right)^{-1}\widehat{\boldsymbol{\Sigma}}_2^{-1}} < \tau(\eta), \quad (25)$$

and finally

$$\|\widehat{\boldsymbol{r}}_1 - \widehat{\boldsymbol{r}}_2\|^2_{\left(\widehat{\boldsymbol{\Sigma}}_1+\widehat{\boldsymbol{\Sigma}}_2\right)^{-1}} < \tau(\eta). \quad (26)$$

We conclude that the constraint is *active* if and only if

$$\|\widehat{\boldsymbol{r}}_1 - \widehat{\boldsymbol{r}}_2\|^2_{\left(\widehat{\boldsymbol{\Sigma}}_1+\widehat{\boldsymbol{\Sigma}}_2\right)^{-1}} \geq \tau(\eta). \quad (27)$$

APPENDIX B
CHOSEN OPTIMIZATION FOR CHANGE LOCALIZATION

In order to optimize the problem (9), we propose to apply the algorithm 1 using IRLS [46] (Iteratively Reweighted Least Squares) to estimate $(\boldsymbol{r}_1^*(\alpha), \boldsymbol{r}_2^*(\alpha))$, and Brent's method [47] for the root-search of α. On the one hand, Brent's method combines classical root-search algorithms (bisection, secant, and inverse quadratic interpolation) to converge faster. On the other hand, we found experimentally that IRLS converges in significantly fewer iterations than a proximal gradient descent or the iPALM algorithm [48] (inertial Proximal Alternating Linearized Minimization), for example. However, IRLS does not provide a sparse solution. Thus, once α is found, we perform a few iPALM steps to obtain a sparse estimate of $|\boldsymbol{r}_1^*(\alpha) - \boldsymbol{r}_2^*(\alpha)|$.

We observe border effects within each patch because border pixels have, on average, fewer correlations with other pixels than the ones in the center (no correlation with pixels outside the patch). Thus, we chose to keep only a 3×3 window in the center of each patch for patch-sizes 5×5 and 7×7 to limit this border effect. This cropping of the estimated solution of the problem for each patch is into account during the calibration of $\tau(\eta)$ described in Appendix C.

APPENDIX C
CALIBRATION OF THE FALSE ALARM PROBABILITY

The probability of false alarm can either be computed at the scale of patches, or at the scale of pixels. Since practitioners are usually more interested in controlling the probability of false alarm at the level of pixels (η), we describe here an automatic method to set the threshold τ applied to detect changed patches, in order to reach this prescribed pixel-level false alarm rate η. Note that the probabilities of false alarm at pixel level and at patch level differ:

$$p(\text{pixel FA}) = \underbrace{p(\text{pixel FA} \mid \text{patch FA})}_{<1} \times p(\text{patch FA}), \quad (28)$$

where "FA" stands for "false alarm".

We build our calibration method defining $\tau(\eta)$ by applying our change detection algorithm only on image pairs without change, *i.e.* under the hypothesis $\mathscr{H}_0$. Such image pairs are readily available thanks to our real/imaginary part decomposition. To calibrate $\tau(\eta)$, we employ a root-search algorithm in order to reach a number of false alarms corresponding to η times the number of pixels.

We can consider two ways to calibrate $\tau(\eta)$: either calibrate across multiple images simultaneously (Fig. 12) or calibrate for each image tested individually. In the first case, we may expect some slight discrepancy between the targeted probability of false alarm and the actual false alarm rate on any given image. In the second case, by definition, η is precisely the false alarm probability for the tested image. However, it is more time-consuming, and enough pixels are necessary to calibrate low false alarm probabilities, since they correspond to only a few pixels (*e.g.* 15 pixels would be erroneously detected as changed under $\mathscr{H}_0$ for $\eta = 10^{-4}$ and an image size of 384×384 pixels). To evaluate both scenarios in the results reported in the article, 5 dates from Saint-Gervais and 4 dates from Altamira were used for TerraSAR-X and Sentinel-1 calibrations, respectively.

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Thomas Bultingaire received the engineering degree in image processing from Télécom Saint-Étienne and the master's degree in Advanced Imaging and Material Appearance (AIMA) from Université Jean-Monnet, Saint-Étienne, France, in 2023. He is currently pursuing a Ph.D. degree with Télécom Paris, Institut Polytechnique de Paris, under the supervision of Florence Tupin, Christophe Kervazo and Loïc Denis. His research focuses on despeckling uncertainty quantification and developing unsupervised change detection techniques using SAR images.



Christophe Kervazo received the Engineering degree from Supélec, Gif-sur-Yvette, France, in 2015, the Master of Science degree from Georgia Institute of Technology, Atlanta, GA, USA, in 2016, and the Ph.D. degree from CEA Saclay, Gif-sur-Yvette, France, in 2019.

He is currently an Associate Professor at Télécom Paris, Palaiseau, France, where he works on imaging applications. His research focuses on interpretable deep learning, both from the neural network architecture point of view and the reliability of their outputs. His applications include remote sensing and medical imaging.



Loïc Denis received the Ph.D. and Habilitation degrees in imaging sciences from the University of Saint-Étienne, Saint-Étienne, France, in 2006 and 2018, respectively.

He is a Full Professor of image processing and artificial intelligence with the University of Saint-Étienne, where he is also the Head of the Optical Design and Image Reconstruction Group, Laboratoire Hubert Curien. He holds an invited position at Image, Data, Signal Division, Telecom Paris's, Palaiseau, France, where he develops artificial intelligence techniques for remote sensing. His research interests include image processing and data science techniques for biomedical imaging, Earth observation, and astronomy, particularly synthetic aperture radar imaging.

Prof. Denis has received several awards for his contributions to the field, including the IEEE ICIP Best Student Paper Award in 2010, the EURASIP EUSIPCO Best Student Paper Award in 2015, the IEEE Geoscience and Remote Sensing Society 2016 Transactions Prize Paper Award in 2016, and the IEEE GRSS Symposium Award in 2022.



Florence Tupin (Senior Member, IEEE) received the B.Eng. and Ph.D. degrees in signal and image processing from the Ecole Nationale Superieure des Telecommunications (ENST), Paris, France, in 1994 and 1997, respectively, and the Habilitation a Diriger des Recherches degree from the University of Rennes, Rennes, France, in 2007.

From 1997 to 1998, she was with SAGEM, Paris, where she worked on fingerprint recognition. She is currently a Professor of image and signal processing with LTCI, Telecom Paris, Palaiseau, France, where she is also the Head of the Image, Data, and Signal Department. She has co-authored more than 200 articles. Her research interests include image processing and analysis, especially for remote sensing and synthetic aperture radar imaging applications, and Earth observation.

Dr. Tupin has been a member of several international and national technical conference committees since 2003. She was an Associate Editor for IEEE Transactions on Geoscience and Remote Sensing from 2007 to 2016. She was a recipient of several awards, among them the IEEE GRSS Transactions Prize Paper Award in 2016, the IEEE GRSS Symposium Prize Paper Award in 2022 and the IEEE GRSS Letter Prize Paper Award in 2026.

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